

SAT UZBEKISTAN

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Numbers Theory

► Easy

1. If x is divisible by both 3 and 4, then the number x must be a multiple of which one of the following?
(A) 8
(B) 12
(C) 15
(D) 18
(E) 21
2. The last digit of the positive even number n equals the last digit of n^2 . Which one of the following could be n ?
(A) 12
(B) 14
(C) 15
(D) 16
(E) 17
3. Which one of the following is divisible by both 2 and 3?
(A) 1005
(B) 1296
(C) 1351
(D) 1406
(E) 1414
4. How many numbers between 100 and 300, inclusive, are multiples of both 5 and 6?
(A) 7
(B) 12
(C) 15
(D) 20
(E) 30
5. Which one of the following equals the product of exactly two prime numbers?
(A) $11 \cdot 6$
(B) $13 \cdot 22$
(C) $14 \cdot 23$
(D) $17 \cdot 21$
(E) $13 \cdot 23$
6. Which one of the following equals the product of the smallest prime number greater than 21 and the largest prime number less than 16?
(A) $13 \cdot 16$
(B) $13 \cdot 29$
(C) $13 \cdot 23$
(D) $15 \cdot 23$
(E) $16 \cdot 21$
7. If m and n are two different prime numbers, then the least common multiple of the two numbers must equal which one of the following?
(A) mn
(B) $m + n$
(C) $m - n$
(D) $m + mn$
(E) $mn + n$

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8. Which one of the following is the first number greater than 200 that is a multiple of both 6 and 8?
- (A) 200
 - (B) 208
 - (C) 212
 - (D) 216
 - (E) 224
9. What is the maximum possible difference between two three-digit numbers each of which is made up of all the digits 1, 2, and 3?
- (A) 156
 - (B) 168
 - (C) 176
 - (D) 196
 - (E) 198
10. The digits of a two-digit number differ by 4 and their squares differ by 40. Which one of the following could be the number?
- (A) 15
 - (B) 26
 - (C) 40
 - (D) 59
 - (E) 73

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➤ **Medium**

11. The number 3 divides a with a result of b and a remainder of 2. The number 3 divides b with a result of 2 and a remainder of 1. What is the value of a ?
- (A) 13
 - (B) 17
 - (C) 21
 - (D) 23
 - (E) 27
12. The remainder when the positive integer m is divided by 7 is x . The remainder when m is divided by 14 is $x + 7$. Which one of the following could m equal?
- (A) 45
 - (B) 53
 - (C) 72
 - (D) 85
 - (E) 100
13. Which one of the following choices does not equal any of the other choices?
- (A) $5.43 + 4.63 - 3.24 - 2.32$
 - (B) $5.53 + 4.73 - 3.34 - 2.42$
 - (C) $5.53 + 4.53 - 3.34 - 2.22$
 - (D) $5.43 + 4.73 - 3.24 - 2.42$
 - (E) $5.43 + 4.73 - 3.14 - 2.22$
14. Each of the two positive integers a and b ends with the digit 2. With which one of the following numbers does $a - b$ end?
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
 - (E) 4

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15. If p and q are two positive integers and $p/q = 1.15$, then p can equal which one of the following?
- (A) 15
 - (B) 18
 - (C) 20
 - (D) 22
 - (E) 23
16. If each of the three nonzero numbers a , b , and c is divisible by 3, then abc must be divisible by which one of the following the numbers?
- (A) 8
 - (B) 27
 - (C) 81
 - (D) 121
 - (E) 159
17. How many positive five-digit numbers can be formed with the digits 0, 3, and 5?
- (A) 14
 - (B) 15
 - (C) 108
 - (D) 162
 - (E) 243
18. If n is a positive integer, which one of the following numbers must have a remainder of 3 when divided by any of the numbers 4, 5, and 6?
- (A) $12n + 3$
 - (B) $24n + 3$
 - (C) $80n + 3$
 - (D) $90n + 2$
 - (E) $120n + 3$
19. x is a two-digit number. The digits of the number differ by 6, and the squares of the digits differ by 60. Which one of the following could x equal?
- (A) 17
 - (B) 28
 - (C) 39
 - (D) 71
 - (E) 93
20. The number 3072 is divisible by both 6 and 8. Which one of the following is the first integer larger than 3072 that is also divisible by both 6 and 8?
- (A) 3078
 - (B) 3084
 - (C) 3086
 - (D) 3090
 - (E) 3096
21. If the least common multiple of m and n is 24, then what is the first integer larger than 3070 that is divisible by both m and n ?
- (A) 3072
 - (B) 3078
 - (C) 3084
 - (D) 3088
 - (E) 3094

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22. a , b , and c are consecutive integers in increasing order of size. If $p = a/5 - b/6$ and $q = b/5 - c/6$, then $q - p =$

(A) $1/60$
(B) $1/30$
(C) $1/12$
(D) $1/6$
(E) $1/5$

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23. A palindrome number is a number that reads the same forward or backward. For example, 787 is a palindrome number. By how much is the first palindrome larger than 233 greater than 233?

(A) 9
(B) 11
(C) 13
(D) 14
(E) 16

24. How many 3-digit numbers do not have an even digit or a zero?

(A) 20
(B) 30
(C) 60
(D) 80
(E) 125

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► Hard

25. Which one of the following is the minimum value of the sum of two integers whose product is 36?

(A) 37
(B) 20
(C) 15
(D) 13
(E) 12

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26. If m and n are two positive integers such that $5m + 7n = 46$, then what is the value of mn ?

(A) 15
(B) 21
(C) 24
(D) 27
(E) 35

27. If a and b are positive integers, and $x = 2 \cdot 3 \cdot 7 \cdot a$, and $y = 2 \cdot 2 \cdot 8 \cdot b$, and the values of both x and y lie between 120 and 130 (not including the two), then $a - b =$

(A) -2
(B) -1
(C) 0
(D) 1
(E) 2

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28. Which one of the following equals the number of multiples of 3 between 102 and 210, inclusive?

(A) 32
(B) 33
(C) 36
(D) 37
(E) 38

29. The sum of the positive integers from 1 through n can be calculated by the formula $n(n + 1)/2$. Which one of the following equals the sum of all the even numbers from 0 through 20, inclusive?
- (A) 50
 - (B) 70
 - (C) 90
 - (D) 110
 - (E) 140
30. a, b, c, d , and e are five consecutive numbers in increasing order of size. Deleting one of the five numbers from the set decreased the sum of the remaining numbers in the set by 20%. Which one of the following numbers was deleted?
- (A) a
 - (B) b
 - (C) c
 - (D) d
 - (E) e
31. A set has exactly five consecutive positive integers starting with 1. What is the percentage decrease in the average of the numbers when the greatest one of the numbers is removed from the set?
- (A) 5
 - (B) 8.5
 - (C) 12.5
 - (D) 15.2
 - (E) 16.66
32. What is the maximum value of m such that 7^m divides into $14!$ evenly?
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
 - (E) 5
33. If $p - 10$ is divisible by 6, then which one of the following must also be divisible by 6?
- (A) p
 - (B) $p - 4$
 - (C) $p + 4$
 - (D) $p - 6$
 - (E) $p + 6$
34. Which one of the following equals the maximum difference between the squares of two single-digit numbers that differ by 4?
- (A) 13
 - (B) 25
 - (C) 45
 - (D) 56
 - (E) 63

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➤ Very Hard

35. $2ab5$ is a four-digit number divisible by 25. If the number formed from the two digits ab is a multiple of 13, then $ab =$

(A) 10
(B) 25
(C) 52
(D) 65
(E) 75

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36. The positive integers m and n leave remainders of 2 and 3, respectively, when divided by 6. $m > n$. What is the remainder when $m - n$ is divided by 6?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

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37. The remainder when $m + n$ is divided by 12 is 8, and the remainder when $m - n$ is divided by 12 is 6. If $m > n$, then what is the remainder when mn divided by 6?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

38. What is the remainder when $7^2 \cdot 8^2$ is divided by 6?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

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39. a, b, c, d , and e are five consecutive integers in increasing order of size. Which one of the following expressions is not odd?

(A) $a + b + c$
(B) $ab + c$
(C) $ab + d$
(D) $ac + d$
(E) $ac + e$

40. How many positive integers less than 500 can be formed using the numbers 1, 2, 3 and 5 for the digits?

(A) 48
(B) 52
(C) 66
(D) 68
(E) 84

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